

# EFFECTS OF HIGH-INTENSITY INTERVAL TRAINING WITH DIFFERENT INTERVAL DURATIONS ON PHYSICAL PERFORMANCE IN HANDBALL PLAYERS

JORGE VIAÑO-SANTASMARINAS, EZEQUIEL REY, SERGIO CARBALLEIRA, AND ALEXIS PADRÓN-CABO

*Faculty of Education and Sport Sciences, University of Vigo, Pontevedra, Spain*

## ABSTRACT

Viaño-Santasmarrinas, J, Rey, E, Carballeira, S, and Padrón-Cabo, A. Effects of high-intensity interval training with different interval durations on physical performance in handball players. *J Strength Cond Res* 32(12): 3398–3406, 2018—The purpose of this study was to compare the effects of 2 high-intensity interval training (HIIT) protocols with different interval durations (short [SI] vs. long [LI]) based on peak running velocity in 30–15 Intermittent Fitness Test ( $V_{IFT}$ ) in handball players. Eighteen highly-trained players (age:  $22.7 \pm 3.9$  years; height:  $181.5 \pm 6.6$  cm; body mass:  $84.7 \pm 14.1$  kg) were randomly assigned to SI (2 sets of 22 intervals of 10-second runs at 95%  $V_{IFT}$ ) or LI (5 sets of 3-minute intervals at 85%  $V_{IFT}$ ) group. The intervention program had to be carried out 2 times a week over 6 weeks. Before and after training period, 10-m sprint, countermovement jump, repeated sprint ability (RSA), and 30–15 $_{IFT}$  were assessed. In the within-group analysis, significant improvements in  $V_{IFT}$  were found in SI (8.18%) and LI (8.19%) from pretest to posttest. Players in both SI and LI also showed significant enhancements in RSA average time (effect size [ES] = 0.72 and 0.38, respectively), total time (ES = 0.72 and 0.38, respectively), and percentage of decrement (ES = 1.08 and 0.77, respectively) from pretest to posttest. Percentage changes in  $V_{IFT}$ , for both groups combined, had a very large correlation with percentage changes in percentage of decrement of RSA ( $r = 0.857$ ). In the between-groups, there were no differences between the training groups (SI and LI) in any variable ( $p > 0.05$ ). These results indicate that both HIIT protocols are effective for fitness development during preseason period in handball players. However, SI should be considered as preferred HIIT method because of its higher specificity.

**KEY WORDS** team sport, jump performance, time-efficient training, sprint

## INTRODUCTION

Handball is a strenuous contact team sport that integrates technical, tactical, and physical skills (16,20,25). Match analysis and physiological monitoring showed that handball involves intermittent high-intensity short-duration activities such as sprinting, jumping, turning, blocking, and throwing (23). Time-motion analysis studies reported that during a 60-minute match, professional handball players cover a total distance of 4,000–5,600 m (24). Analysis of the heart rate, as an indicator of an internal load, has confirmed an elevated intensity during official matches, and players, during most of the time, need to maintain superior effort intensities at 87% of maximal heart rate (24). Additionally, previous studies demonstrated that about 90% of the total energy is driven aerobically (21). Moreover, estimated  $\dot{V}O_2$  assessment during handball games showed that players need at least 71–75% of  $\dot{V}O_{2peak}$  to play at professional level (20). Furthermore, the importance of aerobic capacity in handball is evidenced by its ability to discriminate between players of different levels of performance (8). Collectively, these findings highlight that handball players need high levels of aerobic power that should be carefully considered in talent selection and training process in professional handball (22). Consequently, the development and improvement of this specific physical capacity seem to be crucial to coaches and players.

Considering this background, high-intensity interval training (HIIT) constitutes currently one of the most effective means of improving cardiorespiratory and metabolic function and, in turn, the physical performance (7). High-intensity interval training has been reported to induce greater improvements in aerobic and anaerobic capacity compared with continuous constant-load exercise with half-time of this traditional method (7). Generally, published research in team sports has produced unanimous results with regard to the effectiveness of HIIT on maximal aerobic speed (12), sprint performance (12), lactate threshold (11), and repeated sprint ability (RSA) (8).

According to interval duration, HIIT exercise can be classified as long (3–15 minutes), moderate (1–3 minutes), or short (10 seconds–1 minute) (17), which may induce different central and peripheral adaptations (7). Because of its

Address correspondence to Dr. Ezequiel Rey, [zequirey@uvigo.es](mailto:zequirey@uvigo.es).  
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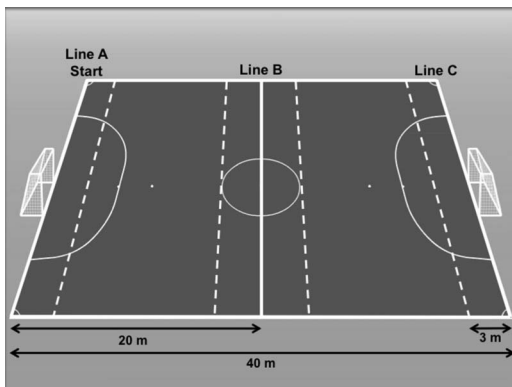
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**TABLE 1.** Physical characteristics (mean  $\pm$  SD) of the players participating in the study.

| Group | N  | Age (y $\pm$ SD) | Body mass (kg $\pm$ SD) | Height (cm $\pm$ SD) |
|-------|----|------------------|-------------------------|----------------------|
| SI    | 10 | 23.0 $\pm$ 4.8   | 83.0 $\pm$ 17.5         | 179.5 $\pm$ 8.0      |
| LI    | 8  | 22.6 $\pm$ 0.3.2 | 86.4 $\pm$ 10.9         | 183.4 $\pm$ 5.0      |

nature, handball, like other team sports, is characterized as an intermittent sport with repeated bouts of short high-intensity activities (2–6 seconds) performed with 6-second intervals (21). Therefore, according to the training specificity principle, it is plausible to assume that HIIT with short intervals may address the specific handball demands and provide more specific training adaptations. Unfortunately, to date, no previous studies have investigated the respective effects of HIIT with short vs. long intervals on physical performance in team sports athletes. Information in this regard could be of practical interest for coaches and strength and conditioning professionals in handball and other team sports.

Therefore, in light of the aforementioned considerations and limitations, the aim of this study was to compare the effectiveness of HIIT with short vs. long intervals administered at an identical work-to-rest ratio (1:1) and with the same training volume per session on selected fitness variables in subelite male handball players. We hypothesized that the use of HIIT with short intervals would provide a more specific training stimulus and improve performance to a greater extent than the HIIT with long intervals. This study was designed to provide coaches and players with relevant information regarding the availability and efficacy of training methods for improving jump, acceleration, RSA, and aerobic performance.



**Figure 1.** Court setup of 30–15 intermittent fitness test.

**METHODS**

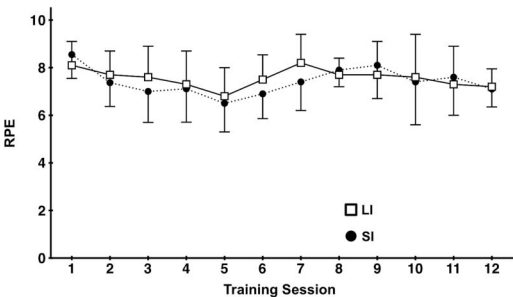
**Experimental Approach to the Problem**

This study used a 2-group, randomized controlled trial design to compare the training effects of 2 different HIIT programs with different interval durations (short [SI] vs. long [LI]). To determine training effects, the following tests were

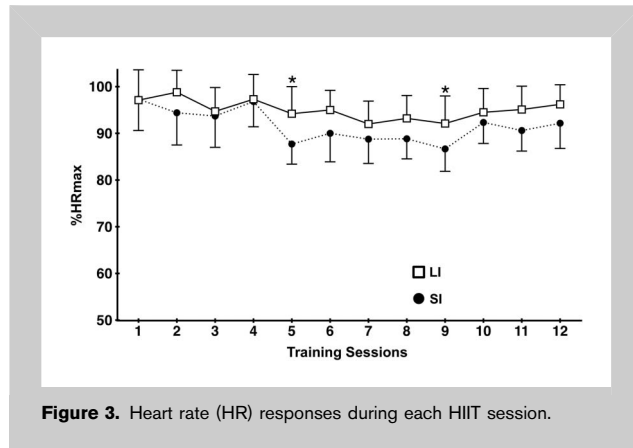
selected: (a) countermovement jump (CMJ), (b) 10-m sprint, (c) RSA test, and (d) 30-15<sub>IFT</sub>. After pretesting, the subjects were randomized by a coauthor not directly involved in testing or the training intervention into one of 2 groups (LI = 8 or SI = 10). We acknowledge that the present study design could have been more powerful with a nonintervention control group. However, the population to draw from in well-trained handball players is limited and dictated the approach taken. The study protocol took place during the preseason period (i.e., August and September), during the sport-specific preparatory phase. The LI and SI groups completed 12 sessions spaced over 6 weeks. The intervention program of each group was added to the usual training routines. To isolate the effects of the 2 training protocols, all subjects completed identical training activities. To reduce the influence of uncontrolled variables, all soccer players were instructed to maintain their habitual lifestyle and normal dietary intake before and during the study.

**Subjects**

Eighteen male amateur handball players (age ranged from 21 to 27 years) participated in this study (subject characteristics are presented in Table 1). With reference to the study of Buchheit et al. (8), an a priori power analysis (13) (G\*Power, version 3.1.9.2, Universität Kiel, Düsseldorf, Germany) with an assumed type I error of 0.05 and a type II error rate of 0.20 (80% statistical power) was conducted for RSA performance



**Figure 2.** Rating of perceived exertion (RPE) after each high-intensity interval training (HIIT) session.



**Figure 3.** Heart rate (HR) responses during each HIIT session.

and revealed that 8 persons per group would be sufficient to observe medium group  $\times$  time interaction effects. All participants were classified as experienced handball players with  $11.2 \pm 5.6$  years of systematic handball training. The players regularly performed 4 weekly soccer sessions with their team and on average exercised  $6.7 \pm 0.9$  h $\cdot$ wk $^{-1}$  in their normal training cycle. The team also regularly competed one official match per week. At the time of the investigation, players were performing 4 weekly training sessions containing plyometric, agility, and handball-specific drills. The players had never participated in a regular/systematic HIIT program. Only players who participated in full training were considered for inclusion. Exclusion criteria were injuries resulting in loss of one or more soccer matches/training sessions in the preceding 3 months before the initiation of the study. The study was approved by the University of Vigo Institutional Review Board for the use of human subjects. In accordance, before participation, the investigator informed all subjects as to the benefits and possible risk associated with the participation in the study and all subjects signed a written informed consent document indicating their voluntary participation.

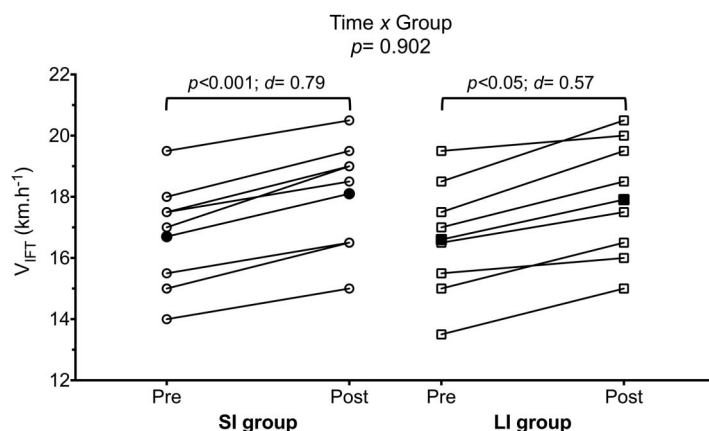
#### Procedures

Players performed 2 testing sessions before and after the 6-week training period. During testing sessions, the participants were required to wear the same athletic equipment and measurements were conducted at the same time of the day to minimize the effect of diurnal variations on the selected parameters during the 2 experimental sessions. All data collection and test sessions were performed in an indoor court where ambient temperature ranged from 18 to 21° C. All tests were performed after 72 hours of rest and at the same venue under identical conditions and supervised by the same test leaders. Before each testing session, players complied with the following pretest guidelines: (a) not to consume any energy/performance-enhancing drinks or supplements 48 hours before testing; (b) not to take beverages containing caffeine or alcohol at least 3 hours before testing; and (c) not to consume food at least 2 hours before testing. Before testing, all participants performed 10 minutes of

**TABLE 2.** Changes in CMJ, 10-m sprint, RSA, and 30-15<sub>IFT</sub> performance following 6 weeks of SI and LI in handball players.\*

|                       | ANOVA $p$ ( $\eta_p^2$ ) |                   |                  |                     |
|-----------------------|--------------------------|-------------------|------------------|---------------------|
|                       | SI ( $n = 10$ )          |                   | LI ( $n = 8$ )   |                     |
|                       | Pre                      | Post              | Pre              | Post                |
| CMJ (cm)              | 31.09 $\pm$ 4.57         | 31.74 $\pm$ 4.13  | 30.95 $\pm$ 6.59 | 31.44 $\pm$ 4.55    |
| 10-m sprint (s)       | 1.78 $\pm$ 0.09          | 1.75 $\pm$ 0.06   | 1.81 $\pm$ 0.10  | 1.80 $\pm$ 0.11     |
| V <sub>IFT</sub> (km) | 16.77 $\pm$ 1.69         | 18.13 $\pm$ 1.74† | 16.58 $\pm$ 2.28 | 17.91 $\pm$ 2.35†   |
| RSA                   |                          |                   |                  |                     |
| AT (s)                | 6.10 $\pm$ 0.26          | 5.94 $\pm$ 0.17†  | 6.18 $\pm$ 0.33  | 6.06 $\pm$ 0.29†    |
| TT (s)                | 36.62 $\pm$ 1.59         | 35.68 $\pm$ 1.07† | 37.08 $\pm$ 2.00 | 36.37 $\pm$ 1.77†   |
| FT (s)                | 5.96 $\pm$ 0.19          | 5.80 $\pm$ 0.18†  | 5.91 $\pm$ 0.19  | 5.89 $\pm$ 0.24     |
| %Dec                  | 2.93 $\pm$ 1.11          | 1.80 $\pm$ 0.97†  | 3.45 $\pm$ 0.85  | 2.85 $\pm$ 0.69†    |
|                       |                          |                   | $\Delta$ (%)     | $\Delta$ (%)        |
|                       |                          |                   | Time             | Time $\times$ group |
|                       |                          |                   | 0.074 (0.186)    | 0.231 (0.088)       |
|                       |                          |                   | 0.472 (0.029)    | 0.899 (0.002)       |
|                       |                          |                   | <0.001 (0.893)   | 0.837 (0.003)       |
|                       |                          |                   | <0.001 (0.555)   | 0.448 (0.036)       |
|                       |                          |                   | <0.001 (0.554)   | 0.448 (0.036)       |
|                       |                          |                   | 0.001 (0.340)    | 0.811 (0.004)       |
|                       |                          |                   | <0.001 (0.648)   | 0.089 (0.170)       |
|                       |                          |                   |                  | 0.540 (0.024)       |
|                       |                          |                   |                  | 0.539 (0.024)       |
|                       |                          |                   |                  | 0.056 (0.210)       |
|                       |                          |                   |                  | 0.107 (0.154)       |

\* $\eta_p^2$  = partial eta squared.  
†Significantly different from pretest ( $p < 0.001$ ).



**Figure 4.** Individual and mean pretesting and posttesting data for the velocity reached at the end of the 30-15<sub>IFT</sub> ( $V_{IFT}$ ) by intervention group (SI = HIIT with short intervals; LI = HIIT with long intervals). Filled circles and squares indicate mean data of the SI and LI, respectively.

standardized warm-up comprising 2 minutes of light active static stretching (10 repetitions for hamstrings, quadriceps, and calf muscles) and 5 minutes of jogging, followed by short-distance accelerations (3 submaximal sprints, progressing to 90% of their maximal velocity for the shuttle distance [30 + 30 m]). This routine was supervised by the team's physical trainer before the tests.

On the first testing session, players performed the following 3 tests:

**Vertical-Jump Performance.** CMJ was performed on a force plate (Ergo Jump Bosco System, Globus, Treviso, Italy) according to the procedures proposed by Bosco et al. (3). Players were allowed 3 trials, with a 1-minute recovery period between. The best trial was used for subsequent analysis. From standing position with the hands fixed on the hips, the players were required to bend their knees to a freely chosen angle and perform a maximal vertical jump as high as possible (26). The hands were held on the hips during the jump to avoid any effect of arm-swing. Participants were instructed to keep their body vertical throughout the jump, avoiding undue lateral and frontal movements, and to land with knees fully extended. Any jump that was perceived to deviate from the required instructions was repeated. The intraclass correlation coefficient (ICC) for test-retest trials was 0.96 (95% confidence interval [CI] 0.93–0.97).

**Ten-meter Sprint Test.** Sprint time was measured by means of a dual infrared reflex photoelectric cell system (DSD Laser System, León, Spain). The photoelectric cells were attached to tripods, raised to a height of 0.9 m, and placed in pairs 1 m apart. All players began from a standing start, with the front foot positioned 0.5 m from the first timing gate, and were

instructed to perform all the sprints with a maximal effort. Players were allowed 3 trials, with a 2-minute recovery period between. The best trial was used for subsequent analysis. During the 2 experimental sessions, the participants were required to wear the same shoes to avoid the effects of different athletic equipment. The ICCs for test-retest trials were 0.96 (95% CI 0.92–0.97).

**Repeated Sprint Ability Test.** Photoelectric cells (DSD Laser System) were used to measure the handball players' performance and to increase test reliability. The RSA protocol was consisted of 6 reps of maximal

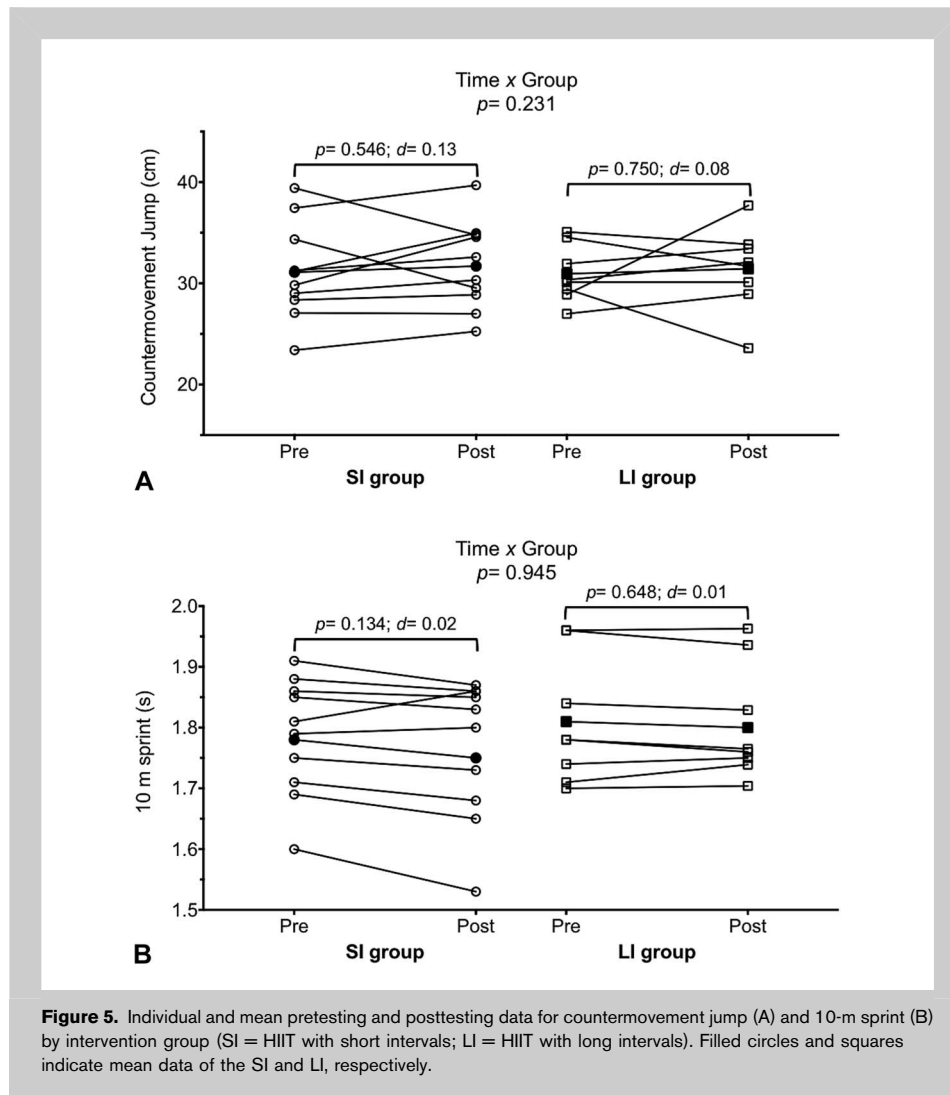
2 × 15 m shuttle sprints. Following each sprint, there was a period of passive standing recovery (20 seconds) (8). Three seconds before starting each sprint, subjects were asked to assume the ready position and await the start signal. The average time (AT), fastest time (FT), and total time (TT) were recorded during the RSA test according to previous studies (15). The percentage of decrement score (%Dec) was then calculated using the following formula, which has been demonstrated as the most valid and reliable method of quantifying fatigue in multiple sprints test (29):

$$\left(100 \times \left(\frac{TT}{\text{ideal sprint time}}\right)\right) - 100$$

where ideal sprint time = 6 × FT.

On the second day, the players performed the following test:

**The 30-15 Intermittent Fitness Test.** The 30-15<sub>IFT</sub> was performed as previously described (4–6). The 30-15<sub>IFT</sub> consisted of 30-second shuttle runs interspersed with 15-second periods of passive recovery periods. The initial running velocity was set at 8  $\text{km} \cdot \text{h}^{-1}$  for the first 30-second run and increased by 0.5  $\text{km} \cdot \text{h}^{-1}$  for every subsequent 45-second stage. Players ran back and forth between 2 lines set 40 m apart at a pace governed by a prerecorded beep. This pacing strategy allowed subjects to run at appropriate intervals and helped them adjust their running speed as they entered into 3-m zones at each end and the middle (20-m line) when a short beep sounds (Figure 1). During the 15-second recovery period, each player walked forward to the closest of the 3 lines (at the middle or at 1 end of the running area, depending on where the previous stage was completed), in preparation for the next stage. Players were instructed to complete



as many stages as possible, and the test ended when a player could no longer maintain the imposed running speed or when they were unable to reach a 3-m zone around each line at the moment of the audio signal on 3 consecutive occasions. If players were unable to complete the stage, then their score was recorded as the stage that they last completed successfully, and the running velocity recorded as their maximal 30-15<sub>IFT</sub> running velocity ( $V_{IFT}$ ) (4). The reliability of  $V_{IFT}$  on 2 consecutive trials separated by 48 hours has been shown to be good (ICC = 0.96; typical error = 0.33 km · h<sup>-1</sup>; 95% CL = 0.26–0.45).

**Heart Rate Measures.** Heart rate (HR) responses were monitored during all the LI and SI training sessions to provide the mean HR percentage (%HR<sub>mean</sub>) and the percentage of maximal HR (%HR<sub>max</sub>). Heart rate was recorded at 1-second intervals (rate to rate) during all HIIT sessions using Polar T<sup>2</sup> system using R-R technology (Polar Electro Oy, Kempele, Finland).

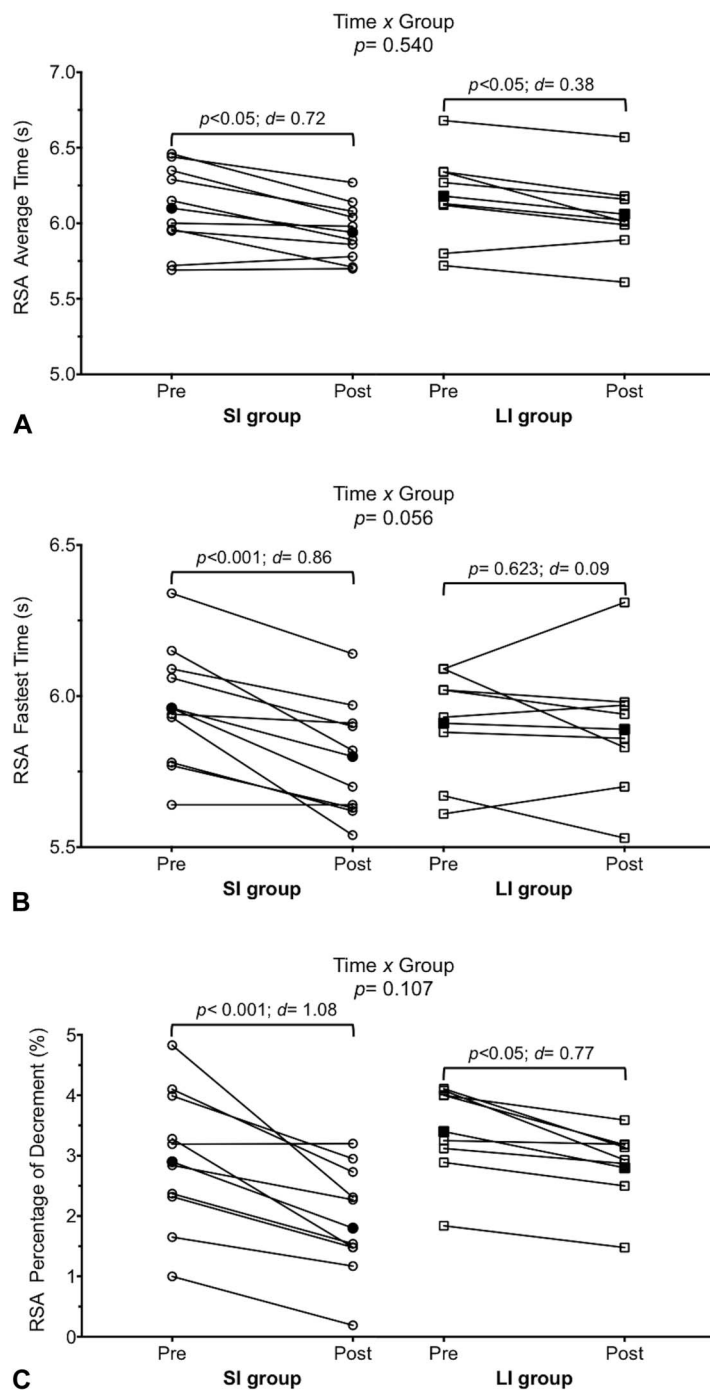
**Rating of Perceived Exertion.** To assess the internal load and the exercise intensity of the training sessions, rating of perceived exertion (RPE) using Foster's 0–10 scale was recorded respectively (14).

**Training Intervention.** After pre-testing, the subjects began one of the 6-week HIIT protocols in addition to the usual handball training. The intervention program had to be carried out 2 times a week (total of 12 sessions) on nonconsecutive days (48-hour rest). Training took places on the same surface as the testing sessions. In both groups, the intensity was individualized according to the  $V_{IFT}$  of each player (5,8). The SI group executed 2 sets of 22 intervals of 10-second runs at 95% $V_{IFT}$  (controlled by the intermittent running distance covered over 10 seconds), interspersed with 10 seconds of passive rest between repetitions and 5 minutes of passive recovery between sets. During the 10-second work period, players had to cover a predetermined distance according to their own  $V_{IFT}$ . After a 10-second passive recovery, play-

ers started running again in the opposite direction. The LI group performed 5 sets of 3-minute intervals at 85%  $V_{IFT}$ , interspersed with 3-minute passive recovery periods. After the fifth training session, the intensity was increased by 3% from the  $V_{IFT}$  to maintain the expected relative intensity. The speed of running was controlled via beep signals from a CD. A certified strength and conditioning specialist supervised all training sessions to ensure that all warm-up activities and sprints were completed with correct technique and with maximum effort.

#### Statistical Analyses

All variables were normally distributed (Shapiro–Wilk test). Data are presented as mean with standard deviation (*SD*). A 2 (group: SI and LI) × 2 (time: pre, post) repeated-measures analysis of variance (ANOVA) were calculated for each parameter. All *p*-values can be seen in table 2. Partial eta squared ( $\eta_p^2$ ) effect sizes for the time × group interaction



**Figure 6.** Individual and mean pretesting and posttesting data for average time (A), fastest time (B), and percentage of decrement (C) in repeated sprint ability (RSA) by intervention group (SI = HIIT with short intervals; LI = HIIT with long intervals). Filled circles and squares indicate mean data of the SI and LI, respectively.

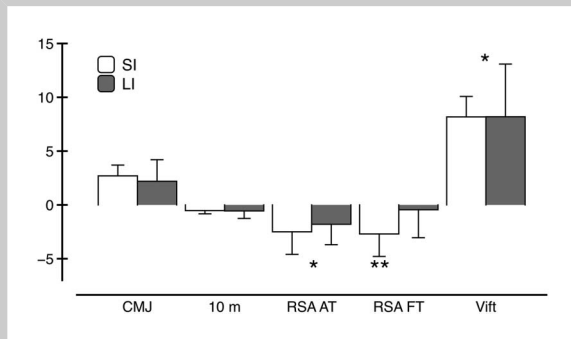
effects were calculated. An effect of  $\eta_p^2 \geq 0.01$  indicates a small,  $\geq 0.059$  a medium, and  $\geq 0.138$  a large effect, respectively (10). Additionally, Cohen's  $d$  effect sizes for identified statistical differences were determined. Effect sizes with values

of 0.2, 0.5, and 0.8 were considered to represent small, medium, and large differences, respectively (10). In addition to this testing, for each variable, percentage difference in the change scores between SI and LI from pretest to posttest was calculated. Correlations between change in  $V_{IFT}$  performance and change RSA performance were assessed by Pearson product-moment correlation coefficient. Magnitude of effect for the correlations was based on the following scale;  $<0.10$  trivial,  $0.10$ – $0.29$  small,  $0.30$ – $0.49$  moderate,  $0.50$ – $0.69$  large,  $0.70$ – $0.89$  very large, and  $>0.90$  nearly perfect (18). Reliability for test-retest trials was assessed using ICC, with a value of  $0.7$ – $0.8$  being questionable and  $>0.9$  being high (28). Significance was set at an  $\alpha$  level of  $0.05$ . All statistical analyses were conducted using the statistical package SPSS for Macintosh (version 21.0, Chicago, IL, USA).

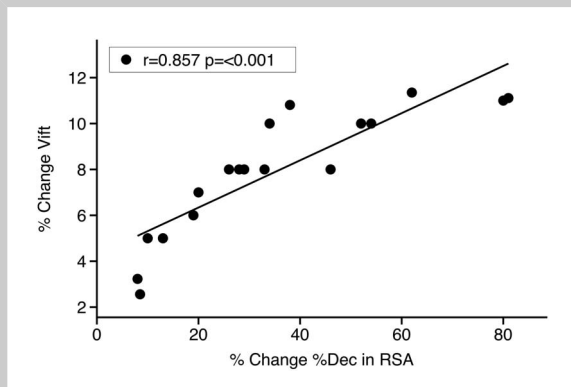
## RESULTS

Rating of perceived exertion was reported throughout the intervention period with a median between 6 and 9 (Figure 2). There were no significant differences in RPE between groups during the training period. Figure 3 shows the HR responses to the HIIT sessions during the study period. Significant differences were observed between groups with relation to the  $\%HR_{max}$  in the fifth and ninth training sessions ( $p \leq 0.05$ ). Absolute values for each parameter at pretest and posttest, together with the ANOVA results, are displayed in Table 2.

In the within-group analysis, significant improvements in  $V_{IFT}$  were found in SI ( $d = 0.79$ ) and LI ( $d = 0.57$ ) from pretest to posttest (Figure 4). There were no significant differences in absolute values for CMJ and 10-m sprint between pretest and posttest in SI and LI groups (Figure 5).



**Figure 7.** Training-induced changes in counter movement jump (CMJ), 10-m sprint, average time during RSA test, fastest time during the RSA test, and velocity reached at the end of the 30-15<sub>IFT</sub> ( $V_{IFT}$ ). \*Significant time effect in both groups ( $p < 0.001$ ). \*\*Significant time effect in SI group ( $p < 0.001$ ).



**Figure 8.** Relationship between percentage changes in velocity reached at the end of the 30-15<sub>IFT</sub> ( $V_{IFT}$ ) and percentage changes in % Dec in RSA test.

Players in both SI and LI also showed significant enhancements in RSA AT ( $d = 0.72$  and  $0.38$ , for SI and LI, respectively), TT ( $d = 0.72$  and  $0.38$ , for SI and LI, respectively), and % Dec ( $d = 1.08$  and  $0.77$ , for SI and LI, respectively) from pretest to posttest (Figure 6). Significant improvements in FT were observed only in SI ( $d = 0.86$ ). Results from the between-group analyses are illustrated in Figure 7. There were no differences between the training groups (SI and LI) in any variable ( $p > 0.05$ ) (Figure 8). Percentage changes in  $V_{IFT}$ , for both groups combined, had a very large correlation with percentage changes in %Dec in RSA ( $r = 0.857$ ; Figure 8).

## DISCUSSION

This study is the first to use specific field test and a controlled study design to compare the effectiveness of 2 HIIT methodologies with different interval durations, i.e., SI and LI, on jump, sprint, aerobic, and RSA performances in adult handball players during the preseason period. The main

findings of this study were that: (a) both SI and LI in addition to normal training during preseason period improved RSA (AT, TT, and %Dec) performance from pretest to posttest; however (b), SI resulted in greater increase in RSA FT compared with LI; and (c)  $V_{IFT}$  significantly improved following 6 weeks in both SI and LI.

It has been well established that explosive actions such as sprinting and jumping are essential fitness components for handball playing (20). In the present study, no significant improvements were observed in 10-m sprint time or CMJ height following 6 weeks of both HIIT programs. These results, which are in agreement with previous data in young handball players (8), may be predictable taking into account that both training programs were specifically designed to improve aerobic capacity (8). However, they are in contrast with those recently reported by Iacono et al. (19) in elite handball players. In the latter, significant changes were found in CMJ and 10-m sprint following 8 weeks of small-sided games training and HIIT, with better results in small-sided games group. Although no electromyographic data were collected in this study, the present results may indicate that the exercise stimuli during both HIIT protocols may not have been high enough to overload the neuromuscular system to substantially improve explosive force production. In this respect, the present data suggest that more specific training strategies (e.g., plyometrics, resisted sprint training, resistance training) than those employed in this study may be required to improve jump and sprint capacities in handball players.

As a result of significant amounts of intermittent sprinting and multidirectional changing movements with brief recovery periods, RSA has been recognized as one of the most relevant physical component in handball (24). Repeated sprint ability is a complex quality related to both neuromuscular (i.e., acceleration and maximal sprint speed, e.g., neural drive or motor unit activation) and metabolic factors (9,15). Thus, training strategies targeting the development of aerobic performance may account for an improvement in RSA (9,15). Different training approaches have shown positive effects on team sports players' RSA. However, although there is good evidence to suggest that aerobic training may be positive for RSA performance, the influence of periodized HIIT with different interval durations on RSA has not been studied yet in team sport players. Results from the present study revealed a significant improvement in RSA (AT, TT, and %Dec) after the training period in SI and LI, highlighting the enhanced ability of the handball players to repeatedly produce maximal sprints and to recover between these sprints. Additionally, SI group also improved RSA FT after 6 weeks of training. The current results are partially in agreement with those of previous investigation in young handball players (8), which had reported beneficial effects of HIIT in RSA performance. Probably, these improvements may be related to a boost in aerobic fitness and muscle buffer capacity that may promote a faster rate of phosphocreatine resynthesis (1,2). This conclusion may be supported by the

association observed between the changes in RSA %Dec and the percentage change in  $V_{IFT}$  (Figure 4). In addition, the improvement in FT performance of the SI may also be explained by the specific changes induced by shuttle-running patterns of SI training. Given the importance of RSA in handball, coaches should consider adding focused HIIT to improve this quality in addition to normal training.

Time-motion analysis of handball match-play has highlighted the intermittent nature of the game (20). Thus, according to the training specificity principle, it is plausible to assume that HIIT with short intervals may recreate the demands of match-play more specifically and provide more specific training adaptations than HIIT with long intervals. By contrast, the present study showed that both HIIT modalities (SI and LI) similarly improved maximal aerobic power, inferred from  $V_{IFT}$  (+8%), during the preparation period in high-level handball players. Buchheit et al. (8) observed similar improvements ( $\approx 7\%$ ) in  $V_{IFT}$  after 10 weeks of in-season HIIT ( $\times 2$  per week of long-interval HIIT) and small-sided games in young handball players. The present data demonstrate that HIIT with short intervals at 105% of  $V_{IFT}$  has no more efficacy than HIIT with long intervals at 85% of  $V_{IFT}$  for the development of aerobic indices in handball players. Despite short intervals HIIT requiring additional muscular actions for decelerations/accelerations, this fact did not induce a greater cardiovascular adaptation in the SI compared with the LI group. The lack of differences between SI and LI may have resulted from the similar cardiorespiratory load imposed by both training methods (27). In a recent study by Tucker et al. (27), physiological responses of long-interval HIIT ( $4 \times 4$  minutes) were compared with short-interval HIIT ( $16 \times 1$  minutes) performed by recreationally active young males. Results revealed relatively similar HR, blood lactate concentration, RPE, and perceived enjoyment over the work bouts during SI vs. LI protocols. In our study, SI and LI elicited fairly similar HR and RPE over the work bouts during all training sessions that likely explain the similar training adaptations observed in both groups. These results have important practical implications and provide further evidence that the  $V_{IFT}$  measured in field setting is a sensitive index for monitoring changes in handball players' cardiorespiratory fitness.

Although the study had many unique aspects (elite-level players), it is important to point out the limitations of the present study. The lack of control group, even taking into account that coaches are reluctant to create a control condition looking to improve the functionality as fast as possible, is a major issue to bear in mind.

In summary, this study demonstrated that the prescription of SI and LI based on  $V_{IFT}$  contributes to improving the RSA and aerobic fitness in handball players. Additionally, the different training protocols used in this study did not negatively affect the jump and sprint performances.

## PRACTICAL APPLICATIONS

The present study was the first to investigate the effects of 2 HIIT protocols with different interval durations on specific

physical attributes of handball players. The results demonstrated that a 6-week period intervention, including either SI or LI twice a week, improve aerobic fitness and the ability to repeat high-intensity efforts with directional changes of adult handball players during the preseason period. Moreover, training effects were independent of the HIIT method, i.e., SI or LI. Thus, coaches are encouraged to include these HIIT methodologies into season planning as a time-efficient approach to improve athletic qualities. Additionally, in light of the findings,  $V_{IFT}$  can be considered as a valid tool for HIIT prescription in handball. These results also show that it is critically important to monitor the player's response to HIIT to ensure that sufficient training intensity is provided. The complementary use of strength training methods (e.g., plyometrics, resistance training) should be considered by coaches as training alternatives to improve jump and sprint performances in handball players.

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